

- 1 (a) The total momentum of a system remains constant provided there is no net external force acting on the system. 1
1
- (b) (i) $-1.20 \text{ kg m s}^{-1}$. 1
- (ii) It is the total momentum of the system which is constant at all points in time. 1
- (c) Taking rightward as positive,
Relative speed of approach $= \frac{2.90}{1.21} - \left(\frac{-4.10}{2.41} \right) = 4.10 \text{ m s}^{-1}$ 1
Relative speed of separation $= \frac{1.50}{2.41} - \left(\frac{-2.70}{1.21} \right) = 2.85 \text{ m s}^{-1}$ 1
Since the two relative speeds are not the same, the collision is not elastic. 1
- (d) Taking rightward as positive,
Average force = Change in momentum of trolley A / time taken
$$= \frac{-2.70 - 2.90}{0.30} = -18.7 \text{ N}$$
 1
Force on trolley A = 18.7 N 1
- (e) Closest (when spring is most compressed) would be when their velocities are the same. 1
Since trolley A is lighter, and since $v = \frac{p}{m}$, the two trolleys will be at their closest before the intersection. 1

[Total: 11]

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

2021 H2 Physics Paper 2 Suggested Solutions

2 (a) Taking moments about X at the wall, and letting the length be l